

Micro-Nano Electro Mechanical System

CL 625



GROUP 6

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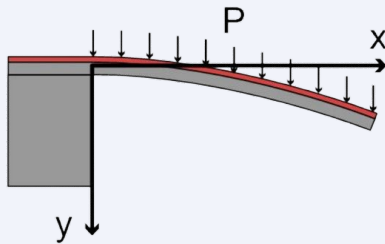
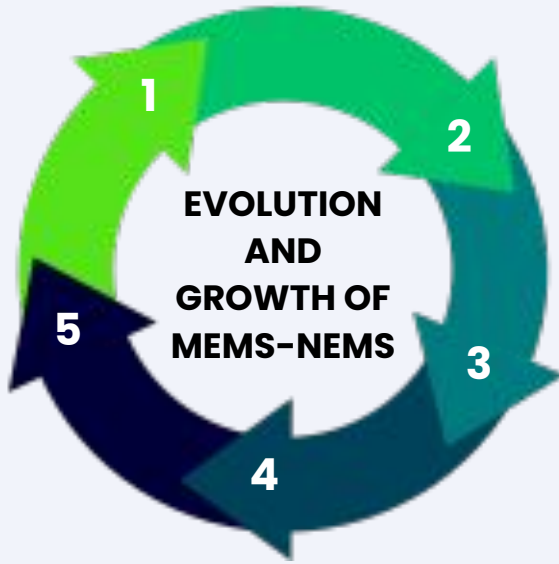
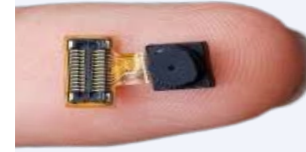


Figure: Cantilever Pressure Sensor

MEMS/NEMS

01

Micro-Nano Electro Mechanical Systems are miniaturized devices fabricated on silicon chips.



CANTILEVER SENSOR

02

Pressure causes a micro-cantilever beam to bend, producing changes in resistance and capacitance.

EVOLUTION

04

Pressure sensing technology has developed from large mechanical instruments to miniaturized micro-nano scale devices.

HISTORY

03

It evolved from mercury barometer (1643) and the Tube gauge (1849), to modern silicon based MEMS.

ADVANCED GROWTH

05

In recent development microfabrication and lithography enable highly miniaturized sensors.

State of Availability, Gap, and Motivation

State

01

Modern **MEMS-based pressure sensors** using piezoresistive and capacitive are commercially available, which provides **high sensitivity and fast response time**.

GAP

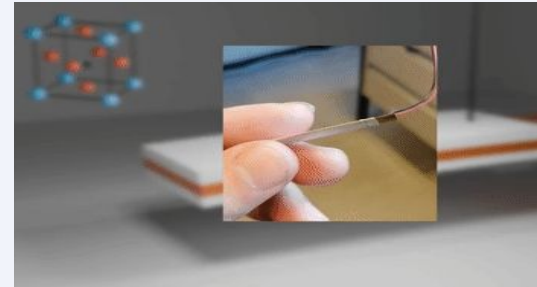
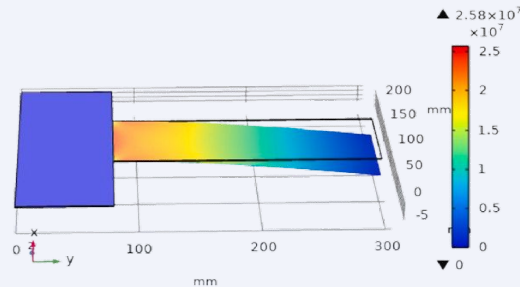
02

Limited **linear sensing range** due to **pull-in instability** in capacitive MEMS sensors.

MOTIVATION

03

By delaying **pull-in instability beyond $g_0/3$** , we maintain a **larger quasi-linear region**, improving measurement accuracy.

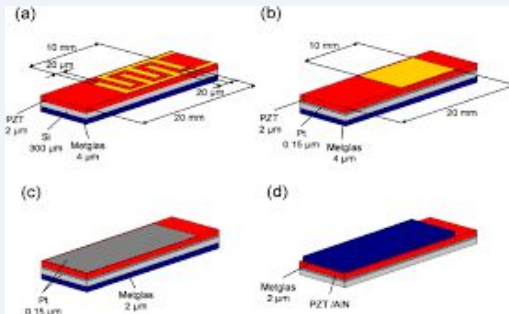


Theoretical Aspects of MEMS

Concept

A cantilever beam forms a **variable capacitor** with a fixed electrode.

External force causes beam deflection → gap change → capacitance change.



Relation

Capacitance relation:

$$C_0 = \epsilon_0 \epsilon_r A / g_0$$

Cantilever tip deflection under force:

$$\delta = FL^3 / 3EI$$

F = force acting at a single point

L = length of the beam

E = modulus of elasticity

I = moment of Inertia

Ideas

Mechanical input → **deflection**

Deflection → **gap change**

Gap change → **measurable capacitance change**

Applications

- Mass sensing
- Bio/chemical adsorption sensing
- Micro-force detection

Capacitance change due to deflection

01

New capacitance becomes

$$C = \epsilon_0 \epsilon_r A / (g_0 - \delta)$$

02

Deflection reduces the capacitor gap:

$$g = g_0 - \delta$$

03

Change in capacitance:

$$\Delta C = \epsilon_0 \epsilon_r A / (g_0 - \delta) - \epsilon_0 \epsilon_r A / g_0$$

04

For MEMS sensors:

$$\delta \ll g_0$$

05

Using small-deflection approximation:

$$\Delta C \approx \epsilon_0 \epsilon_r A \delta / g_0^2$$

Limitation of Parallel Plate MEMS Sensors

- Electrostatic actuation force:

$$F_e = (1/2) * (\epsilon A V^2) / g^2$$

- Mechanical restoring force:

$$F_m = kx$$

- System becomes unstable when:

$$dF_e/dx > dF_m/dx$$

- Pull-in instability occurs at:

$$x_{\text{pull_in}} \approx g_0 / 3$$



Root cause:

Electrostatic force grows rapidly as gap decreases

$$F_e \propto 1 / (g_0 - x)^2$$

Tapered Electrodes to Extend Linear Range

- Proposed geometry:**

Uniform gap:

$$g = g_0 - x$$

Using tapered electrodes:

$$g(y) = g_0 + \alpha y - x$$

Different electrode regions now experience different gaps.

- Total capacitance:**

$$C(x) = \int_0^L (\epsilon w) dy / (g_0 + \alpha y - x)$$

- Solution:**

$$C(x) = (\epsilon w / \alpha) * \ln[(g_0 + \alpha L - x) / (g_0 - x)]$$

Impact on Pull-in and Sensor Performance

- The electrostatic energy stored in the capacitor is:

$$U = (1/2) C(x) V^2$$

where $C(x)$ = displacement dependent capacitance, V = applied voltage

- Electrostatic force is obtained from the derivative of capacitance:

$$F_e = -(1/2) V^2 (dC/dx)$$

- Using the tapered capacitance model:

$$F_e = (1/2) V^2 (\epsilon w / \alpha) [1/(g_0 - x) - 1/(g_0 + \alpha L - x)]$$

This shows that different electrode regions contribute differently to the force.

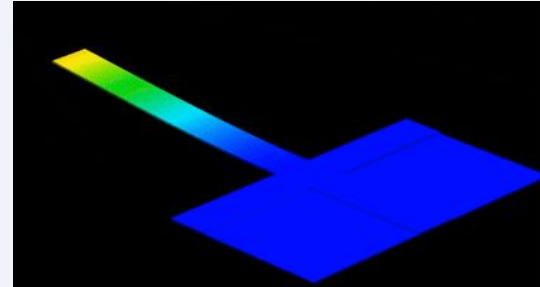
Physical Interpretation

- Large-gap regions reduce overall force growth
- Net electrostatic force increases more gradually

Advantages for MEMS Sensors

- Pull-in instability is delayed
- Pull-in displacement shifts to $\sim 0.45-0.6 g_0$

Result: Higher sensitivity and improved stability.



Device Proposition

A **silicon cantilever** membrane deflects when external pressure is applied.

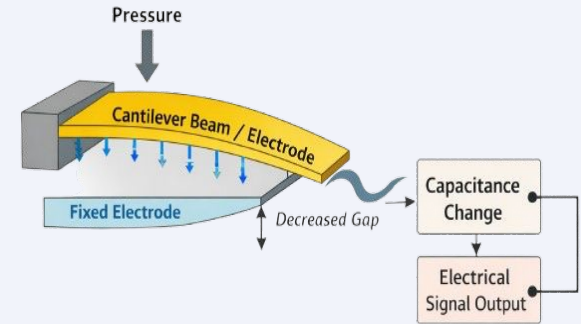
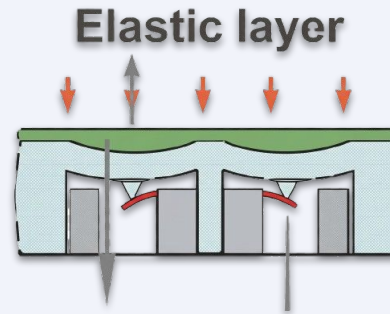
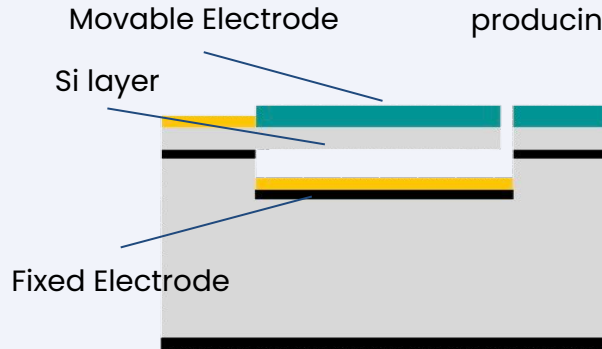
Structural Design:

The proposed sensor mainly consists of:

- A **fixed-free microcantilever beam** fabricated from silicon
- A **pressure-sensitive diaphragm or cavity region**
- **Electrodes layer** for signal detection

The membrane carries a movable electrode, forming a **parallel-plate capacitor with a fixed electrode**.

Pressure-induced displacement changes the gap between the capacitor plates, producing a measurable capacitance variation.



Thank You!